

USER MANUAL

QUESTED - Automated Q&A System for Students



QA Chat with PDFs

Disclaimer: This software uses AI and may sometimes make mistakes. Please verify the answers with the original document.

Text from PDF: "2 Linear Regression.pdf" has been extracted.

what is the linear regression?

the linear regression is a statistical model that is used to predict the probability of a given event occurring . the model is based on the relationship between the observed data and the model . it is also known as logistic regression .

Choose Files 2 Linear Regression.pdf

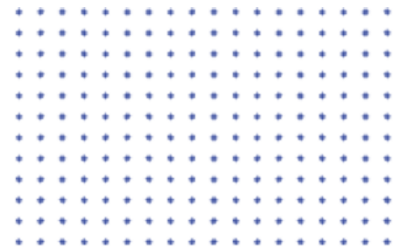
Upload PDFs

Clear All

Ask a question

Ask

INTRODUCTION



Overview

QUESTED is an Automated Q&A System designed to help students find quick and accurate answers from their course materials or textbooks. By utilizing AI and **Natural Language Processing (NLP)**, QUESTED enables students to query specific topics and get relevant answers extracted from predefined academic sources. The goal is to improve learning by providing an interactive and self-paced environment that enhances the accessibility of educational content.

Prerequisites

Before setting up **QUESTED**, you must have the following software installed on your system:

1. **Node.js and npm (Node Package Manager)**

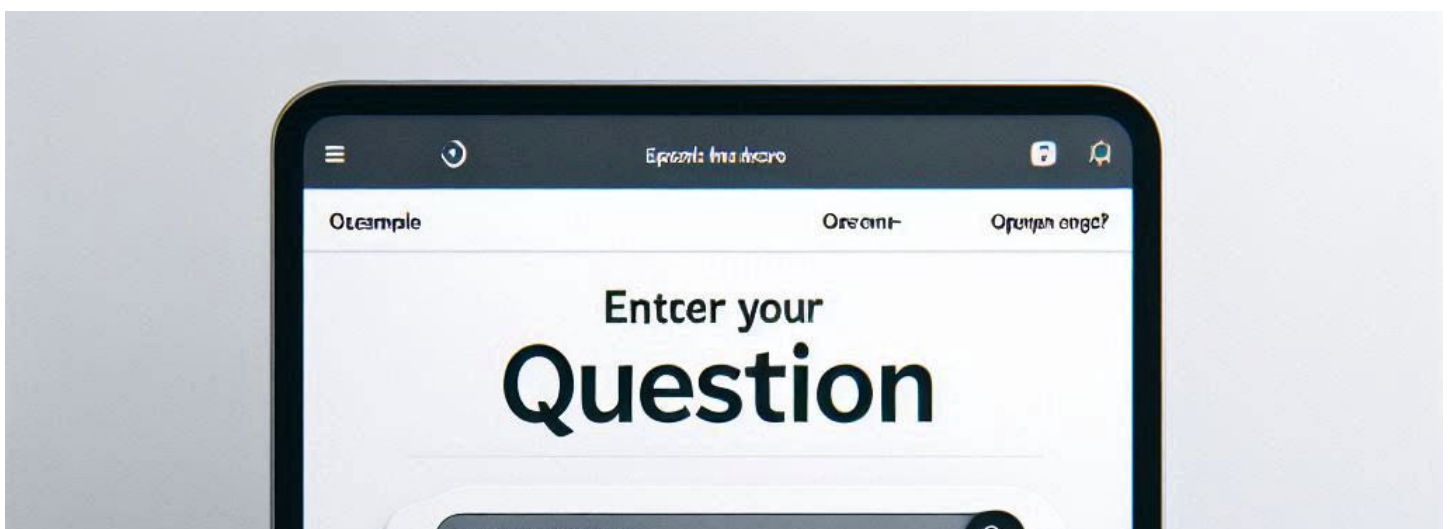
You can download and install it from the official [Node.js website](#).

2. **Python 3.x**

Install Python 3.x from the official [Python website](#).

3. **Tesseract (OCR functionality in Python)**

Tesseract is used for Optical Character Recognition (OCR) to extract text from images. Install it by following the instructions in the [Tesseract OCR installation guide](#).



SYSTEM REQUIREMENTS

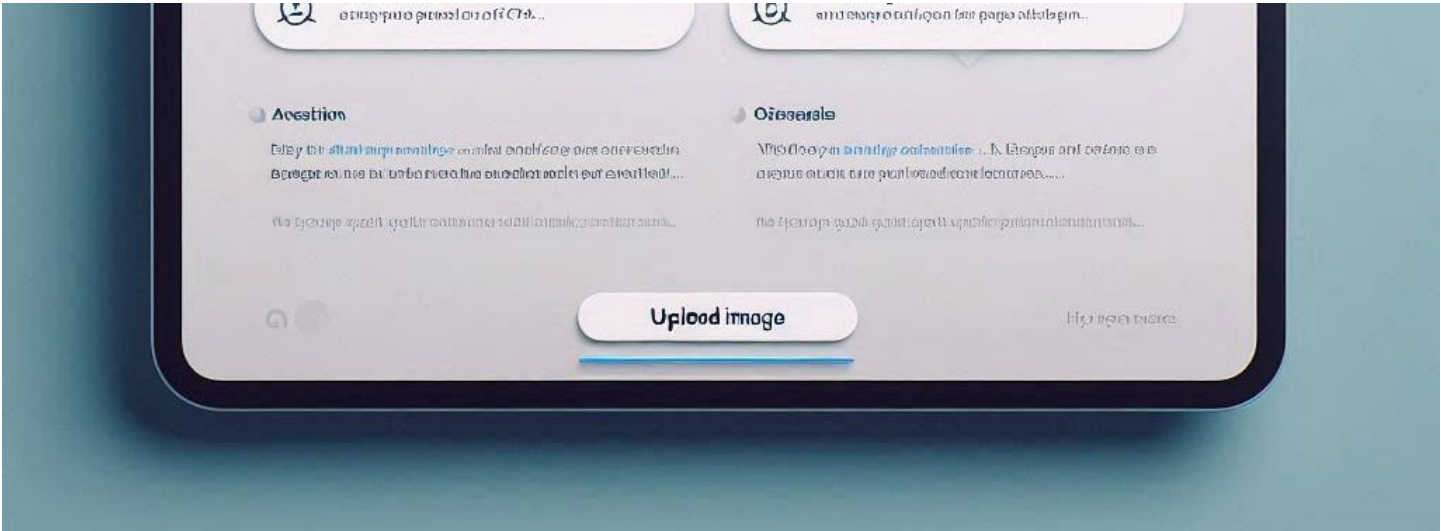
Before installing QUESTED, ensure that your system meets the following minimum requirements:

Hardware Requirements:

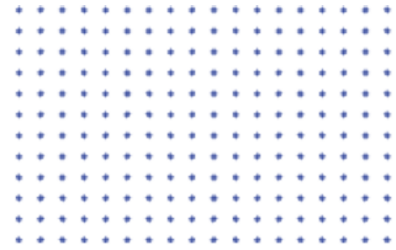
Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5 or higher
RAM	4GB	8GB or higher
Storage	10GB free space	20GB free space
GPU	Not required	Dedicated GPU for faster processing (optional)

Software Requirements:

Software	Required Version
Operating System	Windows 10/11, macOS, Linux
Node.js	v14 or later
npm	v6 or later
Python	v3.8 or later
Tesseract OCR	Latest stable version



PROJECT SETUP AND INSTALLATION GUIDE



1. Clone the Repository

First, clone the QUESTED repository from GitHub:

```
git clone https://huggingface.co/Kav36/quested
```

2. Navigate to the Project Director

After cloning the repository, change into the project directory:

```
cd quested
```

3. Run the Setup Script

To install all required dependencies for both Node.js and Python, run the setup.sh script:

```
./setup.sh
```

This will automatically install:

- Node.js dependencies via npm install
- Python dependencies via pip install

Once the script completes, the system will be ready to run.

RUNNING THE APPLICATION

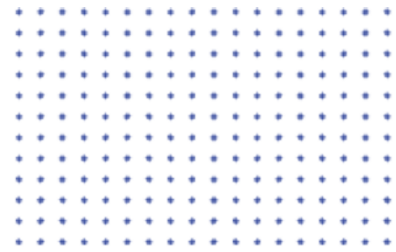
1. Start the Node.js Server

Once the setup is complete, you can start the **Node.js server** by running the following command:

```
npm start
```

This will launch the Node.js server, which will serve as the backend for the Q&A system.

INTERFACE OVERVIEW



The **QUESTED** system provides an easy-to-use web interface designed to help students interact with the Q&A system efficiently.

Key Interface Components:

Welcome page:

The welcome page will display a 'Start Chatting' button, allowing students to begin their interaction with the Q&A system.

Homepage:

The homepage will display a search box where students can type in their questions related to their course materials or textbooks.

Upload PDF:

The homepage will display an Upload PDF button, allowing students to upload course materials, textbooks, or notes.

Search Input:

A simple text input field where students can type their questions (e.g., "What is photosynthesis?").

Result Area:

After submitting a question, the answer will be displayed here (like chat), showing the most relevant response extracted from predefined academic sources.

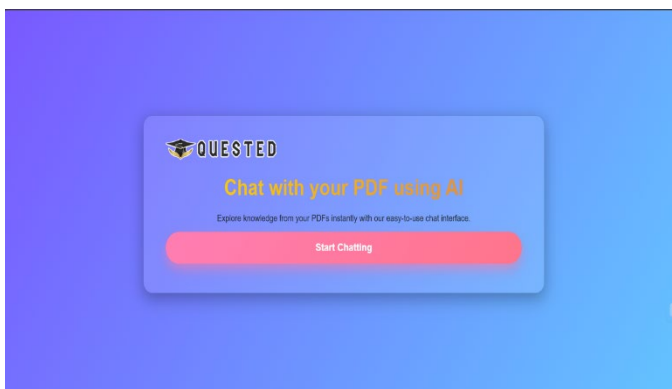


Fig1. Welcome Screen of QUESTED

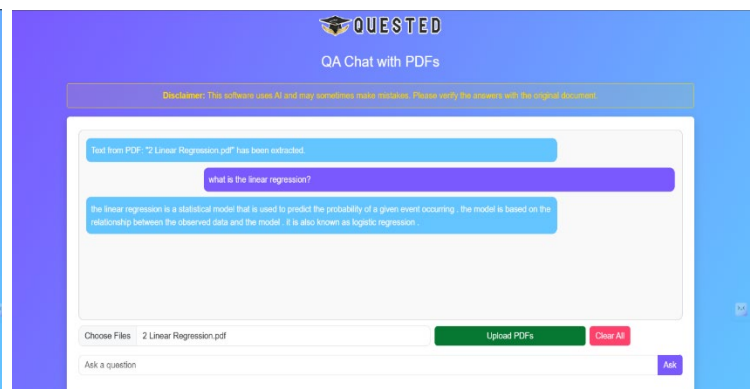


Fig2. Home Screen of QUESTED

USING QUESTED

1. Upload a PDF Document

- Click the **"Choose Files"** button to select a PDF document from your local storage.
- The selected file's name will appear next to the button.
- Click the **green "Upload PDFs"** button to begin extraction.
- Once the extraction is complete, a confirmation message will appear:

"Text from PDF: '[filename].pdf' has been extracted."

2. Ask a Question

- After uploading a PDF, the chat interface will be available.
- In the **"Ask a question"** input field, type your question (e.g., "What is photosynthesis?").
- Click the **"Ask"** button to submit your query.
- Questions appear in a **purple shade**, while user input text appears in **blue**.

3. View Results

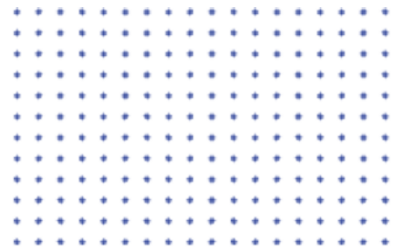
- The system processes the question and provides an answer extracted from the uploaded document.
- The response may include relevant context, references, or citations.
- A disclaimer at the top of the chat reminds users:

"This software uses AI and may sometimes make mistakes. Please verify the answers with the original document."

4. Manage Uploaded Files

- Users can upload more PDFs if needed.
- To remove all files and reset the chat history, click the **red "Clear All"** button.
- This will clear all uploaded documents and restart the session.

TROUBLESHOOTING



Common Issues:

Error: Node.js or Python dependency not installed

Ensure that you ran the setup.sh script correctly. Check for any installation errors, and make sure both Node.js and Python dependencies are installed properly.

OCR not working

Make sure that Tesseract OCR is installed and configured properly. Refer to the Tesseract installation guide for troubleshooting tips.

The system is not responding

Ensure the Node.js server is running. If it's not, restart the server with npm start

VERSION HISTORY

Version	Release Date	Updates & Features
1.0.0	February 2025	Initial Release: AI-powered Q&A, OCR support.

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